

# Statistics & Data Analysis

Comprehensive Midterm Cheat Sheet · Probability → Random Variables → Distributions → Normal & CLT

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Classes 01–04

**How to use:** Formulas sit in blue boxes; **METHOD** notes in green; **TRAP** = the mistakes that cost points. Every rule is cross-checked against the sample-midterm & HW solution keys. The fully-worked Sample Midterm is §9.

## Probability

Conditional, Bayes, unions, total prob., independence, tables

## Random Variables

PMF/CDF,  $E(X)$ , Var,  $\sigma$ ;  $E(XY)$ ; rules for  $aX+bY$

## Distributions

Binomial, Poisson, Hypergeometric, Covariance/Correlation

## Normal & CLT

z-scores, z-table, sum vs. mean, normal=binomial

## §1 Probability Foundations

### Vocabulary

- Experiment** → process with chance outcomes. **Sample space (S)** = set of *all* outcomes. **Event** = a subset of outcomes (capital letters A, B, ...).
- Simple events** = the most basic, mutually-exclusive outcomes; S is the collection of all of them. Ex: two coins →  $S = \{HH, HT, TH, TT\}$ .
- Mutually exclusive** = both cannot happen at once ( $A \cap B = \emptyset$ ).
- Intersection  $A \cap B$**  = both occur. **Union  $A \cup B$**  = A or B or both.

### The Core Probability Rules

#### RANGE & TOTAL

$$0 \leq P(A) \leq 1 \cdot P(S) = 1 \cdot \sum p(\text{all outcomes}) = 1$$

#### COMPLEMENT

$$P(A) = 1 - P(\bar{A})$$

#### EQUALLY-LIKELY OUTCOMES

$$P(A) = m / M \quad (m \text{ favorable of } M \text{ equally-likely})$$

#### ADDITION (GENERAL)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

#### ADDITION (MUTUALLY EXCL.)

$$P(A \cup B) = P(A) + P(B)$$

#### MULTIPLICATION

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

#### INDEPENDENT EVENTS

$$P(A \cap B) = P(A) \cdot P(B)$$

### Conditional Probability & Independence

#### CONDITIONAL

$$P(A|B) = P(A \cap B) / P(B)$$

**Independence test** — A, B independent iff any one holds:

$$P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B) \Leftrightarrow P(A \cap B) = P(A)P(B)$$

#### METHOD

To decide independence: compute  $P(A) \cdot P(B)$  and compare to the given  $P(A \cap B)$ . Equal → independent; not equal → dependent.

#### TRAP

**Mutually exclusive ≠ independent.** If A, B are mutually exclusive with  $P > 0$  they are *dependent* (knowing A happened forces  $P(B)=0$ ). Conditioning *shrinks* the sample space.

### Total Probability & Bayes' Theorem

#### LAW OF TOTAL PROBABILITY

$$P(A) = P(A|B) \cdot P(B) + P(A|\bar{B}) \cdot P(\bar{B})$$

#### BAYES' THEOREM

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|\bar{B}) \cdot P(\bar{B})}$$

#### METHOD

**Two-stage / "jar" & testing problems:** Stage 1 = which path (die roll, coin, jar). Stage 2 = outcome given the path. Multiply along a branch; add branches for total probability; Bayes to reverse (find the cause given the effect). A **tree** or a **frequency table scaled to 10,000** both work.

#### EX · DISEASE TEST (BAYES)

$P(\text{Dis}) = .01$ ,  $P(\text{Pos}|\text{Dis}) = .98$ ,  $P(\text{Neg}|\bar{\text{Dis}}) = .99 \Rightarrow P(\text{Pos}|\bar{\text{Dis}}) = .01$ . Find  $P(\text{Dis}|\text{Pos})$ .

$$P(\text{Dis}|\text{Pos}) = (.98)(.01) / [(.98)(.01) + (.01)(.99)] = .0098 / .0197 = \approx .498$$

Table of 10,000: 98 true-pos, 99 false-pos →  $98/197 = .498$ .

### Contingency (Two-Way) Tables

**Joint** = one cell ÷ grand total. **Marginal** = row/col total ÷ grand total. **Conditional** = cell ÷ that row-or-column total (the "given" resets the denominator).

|        | USA | Non-US | Total |
|--------|-----|--------|-------|
| Male   | 22  | 18     | 40    |
| Female | 13  | 9      | 22    |
| Total  | 35  | 27     | 62    |

$P(\text{Male}|\text{Non-US}) = (40+27-18)/62 = 49/62$ .  $P(\text{Asian}|\text{Female}) = \text{cell}/\text{row} = 3/22$ .  $P(\text{Male}|\text{not-Euro}) = 35/54$ .

## §2 Counting: Factorials, Permutations, Combinations

#### FACTORIAL

$$x! = x(x-1) \cdots (2)(1) \cdot 0! = 1$$

#### PERMUTATIONS — ORDER MATTERS

$${}_n P_k = n! / (n-k)!$$

#### COMBINATIONS — ORDER DOESN'T

$$C(n,k) = {}^n C_k = n! / [k!(n-k)!]$$

- Order x distinct items in a row →  $x!$

- Pres/VP/Treas from 10 (order matters) →  $10 \cdot 9 \cdot 8 = 720 = 10!/7!$
- Committee of 3 from 8 (no order) →  $C(8,3) = 56$
- Choose 2 of 5 firms,  $P(\text{they're A\&B}) = 1/C(5,2) = 1/10$ ;  $P(\text{E among 3 chosen of 5}) = C(4,2)/C(5,3) = 6/10 = 3/5$

#### METHOD

"Arrange / rank / assign titles" → permutation. "Choose / select / committee / sample" → combination.

## §3 Random Variables: PMF, E(X), Variance

### Definitions

- Random variable (X)** = takes numeric values by chance. **Discrete** = countable ("counting"); **Continuous** = measured, in-between values.
- PMF / pdf:**  $p(x)=P(X=x)$ . Requires  $0 \leq p(x) \leq 1$  and  $\sum p(x) = 1$ .
- CDF:**  $F(x)=P(X \leq x)$ . Then  $P(X > x) = 1 - F(x)$ ;  $P(a \leq X \leq b) = F(b) - F(a-1)$  for integers. Get pmf from cdf by differencing.

### Expected Value & Variance

#### MEAN / EXPECTED VALUE

$$\mu_X = E(X) = \sum x \cdot p(x)$$

#### VARIANCE — DEFINITION

$$\sigma_X^2 = \text{Var}(X) = \sum (x - \mu)^2 \cdot p(x)$$

#### VARIANCE — SHORTCUT (USE THIS!)

$$\text{Var}(X) = \sum x^2 \cdot p(x) - \mu^2 = E(X^2) - [E(X)]^2$$

#### STANDARD DEVIATION

$$\sigma_X = \sqrt{\text{Var}(X)} \quad (\text{always } \geq 0; \text{ in original units})$$

**METHOD** Build a table: columns  $x$ ,  $p(x)$ ,  $x \cdot p(x)$ ,  $x^2 \cdot p(x)$ . Sum col 3  $\rightarrow E(X)$ . Sum col 4  $\rightarrow E(X^2)$ .  $\text{Var} = E(X^2) - [E(X)]^2$ .

### Rules for E and Var (a, b constants)

$$E(a) = a$$

$$\text{Var}(a) = 0$$

$$E(aX+b) = a \cdot E(X) + b$$

$$\text{Var}(aX+b) = a^2 \cdot \text{Var}(X)$$

$$E(X+Y) = E(X)+E(Y) \text{ always}$$

$$\sigma_{aX} = |a| \cdot \sigma_X$$

**TRAP** Adding a constant **b** shifts the mean but **NOT** the variance ( $\text{Var}(X+b)=\text{Var}(X)$ ). A multiplier **a** comes out squared in variance ( $a^2$ ) but plain in the mean.

**METHOD · EMPIRICAL RULE FOR RVS**  $\approx 3/4$  of values within  $\mu \pm \sigma$ ;  $\approx 95\%$  within  $\mu \pm 2\sigma$ .

## §4 Covariance, Correlation & Sums of Random Variables

#### COVARIANCE

$$\text{Cov}(X,Y) = E(XY) - \mu_X \mu_Y = E[(X - \mu_X)(Y - \mu_Y)]$$

#### CORRELATION ( $-1 \leq \rho \leq 1$ )

$$\rho_{XY} = \text{Cov}(X,Y) / (\sigma_X \cdot \sigma_Y) \Rightarrow \text{Cov}(X,Y) = \rho_{XY} \cdot \sigma_X \cdot \sigma_Y$$

#### $E(XY)$

$$E(XY) = \sum (x \cdot y) \cdot P(x,y) \quad \text{sum over joint outcomes}$$

### Variance of a Sum / Combination

#### TWO VARIABLES

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X,Y)$$

#### DIFFERENCE

$$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) - 2 \cdot \text{Cov}(X,Y)$$

#### GENERAL WEIGHTED

$$\text{Var}(aX+bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \cdot \text{Cov}(X,Y)$$

#### SCALING COVARIANCE

$$\text{Cov}(aX,bY) = ab \cdot \text{Cov}(X,Y) \quad \text{Cov}(X,X) = \text{Var}(X)$$

**INDEPENDENT  $\Rightarrow$  COV = 0** Then  $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$  and  $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$  too. **Variances add; standard deviations do NOT** ( $\sigma_{X+Y} \neq \sigma_X + \sigma_Y$ ).

### EX · COVARIANCE DRILL

$$\sigma_X=10, \sigma_Y=12, \sigma_Z=20; \rho_{XY}=.8, \rho_{XZ}=-.5.$$

$$\text{Cov}(X,Y) = .8 \cdot 10 \cdot 12 = \mathbf{96}; \text{Cov}(X,Z) = -.5 \cdot 10 \cdot 20 = \mathbf{-100}; \text{Cov}(2X,3Y) = 6 \cdot 96 = \mathbf{576}.$$

$$\text{Var}(X+Y) = 100 + 144 + 2(96) = \mathbf{436}. \text{Var}(3X+5Z) = 9(100) + 25(400) + 2 \cdot 15(-100) = \mathbf{7900}.$$

### EX · PORTFOLIO (WEIGHTS SUM TO 1)

$$\text{Return} = w_1 X + w_2 Y. w = .4/.6, E = .12/.19, \sigma = .09/.28, \rho = -.4.$$

$$E(R) = .4(.12) + .6(.19) = \mathbf{.162}$$

$$\text{Var} = .4^2(.09^2) + .6^2(.28^2) + 2(.4)(.6)(-.4)(.09)(.28) = .0247 \rightarrow \sigma = \mathbf{.157}$$

Negative  $\rho$  lowers portfolio risk — the diversification effect.

## §5 The Named Discrete Distributions

### Binomial (n, $\pi$ ) — fixed # of independent trials

**Use when:**  $n$  fixed trials · each success/fail · constant  $\pi$  · independent (**sampling WITH replacement**).  $X = \#$  successes  $= Y_1 + \dots + Y_n$ , a sum of  $n$  **Bernoulli** trials (one trial:  $E = \pi$ ,  $\text{Var} = \pi(1-\pi)$ ).

$$P(X=k) = C(n,k) \cdot \pi^k \cdot (1-\pi)^{n-k}$$

$$E(X) = n\pi \quad \text{Var}(X) = n\pi(1-\pi) \quad \sigma = \sqrt{n\pi(1-\pi)}$$

**"At least one"**  $= 1 - P(0) = 1 - (1-\pi)^n$ . Excel: =BINOM.DIST(k, n,  $\pi$ , FALSE) for  $P(X=k)$ ; TRUE for  $P(X \leq k)$ .

### EX · SALESPERSON, N=25 VISITS, $\pi=.3$

$$P(8 \text{ sales}) = C(25,8) \cdot (.3)^8 \cdot (.7)^{17} = \mathbf{.165}. E = 25 \cdot (.3) = \mathbf{7.5}; \sigma = \sqrt{25 \cdot .3 \cdot .7} = \mathbf{2.29}.$$

Poisson (λ) — random events over time/space

Use when: counting events that occur one-at-a-time, independently, at a steady rate; no upper limit on x.

$P(X=k) = e^{-\lambda} \cdot \lambda^k / k! \quad (k=0,1,2,\dots)$

$E(X) = \text{Var}(X) = \lambda \cdot \lambda = \text{rate} \times \text{duration}$

**METHOD** Scale λ to the window asked: 4/hr over 2 hr → λ=8; over 15 min → λ=1. "At least one" = 1 - e<sup>-λ</sup>.

**EX · CALLS AT 4/HR (POISSON)**  
P(5 in 2 hr): λ=8 → e<sup>-8</sup>·8<sup>5</sup>/5!=**.0916** . P(≥1 in 1 hr): λ=4 → 1-e<sup>-4</sup>=**.982** .

Hypergeometric (N, r, n) — sampling WITHOUT replacement

$P(X=x) = \frac{C(r,x) \cdot C(N-r, n-x)}{C(N,n)}$

N=population, r=successes in pop, n=sample, x=successes in sample. **Without replacement** (draws not independent) → hypergeometric, not binomial.

§6 The Normal Distribution & z-Scores

Properties

- Continuous, **bell-shaped, symmetric** about μ. Two parameters: mean μ, variance σ<sup>2</sup>.
- Only the **number of SDs from the mean** matters — that's what standardizing captures.
- Continuous ⇒ P(X=a)=0, so **< and ≤ are interchangeable**.

**STANDARDIZE**  
 $Z = (X - \mu) / \sigma \rightarrow Z \sim \text{Normal}(0, 1)$

**INTERVAL**  
 $P(a \leq X \leq b) = P(X \leq b) - P(X \leq a)$

**"FIND THE VALUE" / PERCENTILE**  
 $x = \mu + z \cdot \sigma$  (z from the target tail area)

Continuous RV basics

Probability = **area under the density curve**; any single point has probability 0. Notation **X ~ N(μ, σ<sup>2</sup>)** (2nd parameter is the *variance*). **Uniform** on [1,5]: height = 1/(5-1) = ¼ → P(2≤X≤4)=2·¼=½.

EX · 20 MARBLES (12 RED), DRAW 5, NO REPLACEMENT

P(exactly 3 red)=C(12,3)·C(8,2)/C(20,5)=**.397** (with replacement → Binomial: C(5,3)·6<sup>3</sup>·4<sup>2</sup>=.346).

Which distribution? — quick selector

| Situation                                  | Model        |
|--------------------------------------------|--------------|
| Fixed n trials, constant p, w/ replacement | Binomial     |
| Draw without replacement from finite pop   | Hypergeom.   |
| Events per time/area, steady rate          | Poisson      |
| Sum/mean of many RVs; large n              | Normal (CLT) |

**APPROX** Hypergeometric ≈ Binomial when n is small vs. N (pop large): P(none)≈(1-r/N)<sup>n</sup>.

Reading the class z-table

This course's table gives **A(z)=P(0≤Z≤z)** (area from the middle out). Convert:

- P(Z≤z) = **0.5 + A(z)** (z>0)
- P(Z>z) = **0.5 - A(z)**
- P(Z≤-z) = 0.5 - A(z) (**symmetry**)
- P(-a≤Z≤b) = A(a)+A(b)

**METHOD** Always draw the bell, mark μ, shade what's asked, then translate to A(z) pieces. "Find k with P(0≤Z≤k)=.40" → read the table backwards → k≈1.28.

Some tables (& Excel NORM.S.DIST) give the full cumulative Φ(z)=P(Z≤z) directly — then read it straight off, no +.5.

**INTERPOLATE** Between rows: z ≈ z<sub>lo</sub> + [(target-A<sub>lo</sub>)/(A<sub>hi</sub>-A<sub>lo</sub>)]·(z<sub>hi</sub>-z<sub>lo</sub>). Ex: Φ=.90 lies between .8997 (1.28) & .9015 (1.29) → k≈1.282.

**EX · X ~ N(100, 10<sup>2</sup>)**  
P(92<X<104)=P(-.8<Z<.4)=Φ(.4)-Φ(-.8)=.6554-.2119=**.4435**

**EX · COKE DEMAND ~ N(150, 20)**  
P(X≤200): z=(200-150)/20=2.5 → .5+.4938=**.9938**  
P(X≥180): z=1.5 → .5-.4332=**.0668**  
Stock so P(run out)=.1: z=1.28 → 150+1.28(20)=175.6 → stock **176** .

§7 Sampling Distributions & the Central Limit Theorem

**CLT:** the **sum** (or **average**) of many independent, identically-distributed RVs is ≈ **Normal** — even when the underlying variable is not normal. Conditions: iid + independent + large n (**rule of thumb n ≥ 30**).

| SUM OF N            | AVERAGE OF N ( $\bar{X}$ )  |
|---------------------|-----------------------------|
| mean = nμ           | mean = μ                    |
| SD = σ√n            | SE = σ/√n                   |
| Z = (T - nμ)/(σ/√n) | Z = ( $\bar{X}$ - μ)/(σ/√n) |

**TRAP** **Sum vs. average.** Sum spreads out (σ√n); the average tightens (σ/√n). Pick the right one and don't mix variances with SDs.

Two special cases

- Underlying is Normal** → the sum/average is **exactly** Normal for any n (not an approximation).
- Normal = Binomial:** valid when **nπ > 5 and n(1-π) > 5**; then X ≈ Normal with mean nπ, variance nπ(1-π).

**THE CLASSIC EXAM TRAP** "n>30 ⇒ an individual observation is normal" is **FALSE**. The CLT normalizes the **sample mean/sum** only. A single item keeps the population's shape no matter how big n is.

**EX · CARS SOLD/DAY: E=1.7, VAR=.81 (SUM)**  
300 days: sum mean=300(1.7)=510, SD=√(300·.81)=15.59.  
P(fewer than 495)= z=(495-510)/15.59=-.96 → **≈ .17**

**EX · BRANCH EMPLOYEES M=50, Σ=15 (AVERAGE)**  
n=40: SE=15/√40=2.37 → z=(55-50)/2.37=2.11 → P( $\bar{X}$ >55)=**.017**  
n=100: SE=15/√100=1.5 → z=3.33 → P( $\bar{X}$ >55)=**.0004**  
Bigger n → smaller SE → tighter around μ. Assumes iid, n large.

## §8 Descriptive Statistics & Data (Class 01)

### Data & sampling vocabulary

- Quantitative** (numbers/arithmetic) vs **Qualitative** (categories). **Discrete** (count) vs **Continuous** (measure).
- Nominal** (no order) vs **Ordinal** (ordered, e.g. Likert). **Cross-sectional** vs **Time series**. **Univariate** vs **Multivariate**.
- Random sample**: every element equally likely to be chosen.

**KEY IDEA** **Parameter** = summary of the **population** (FIXED). **Statistic** = summary of the **sample** (RANDOM). Populations are fixed; statistics vary sample-to-sample.

### Center & spread

SAMPLE / POP MEAN  
 $\bar{x} = \sum x/n$  ·  $\mu = \sum x/N$

MEDIAN  
 middle; even  $n \rightarrow$  avg of 2

POPULATION VARIANCE  
 $\sigma^2 = \sum (x - \mu)^2 / N = (\sum x^2 - N\mu^2) / N$

SAMPLE VARIANCE — DIVIDE BY  $N-1$   
 $s^2 = \sum (x - \bar{x})^2 / (n-1) = (\sum x^2 - n\bar{x}^2) / (n-1)$

**WHY  $N-1$ ?** So  $s^2$  is an **unbiased estimator** of  $\sigma^2$ . Population  $\sigma^2$  uses  $N$  and  $\mu$ ; sample  $s^2$  uses  $n-1$  and  $\bar{x}$ . Mixing these up is the classic error.

### z-scores, quartiles & outliers

$$z = (x - \mu) / \sigma \text{ or } (x - \bar{x}) / s$$

- Outlier by z**:  $|z| \geq 3$  outlier;  $2 \leq |z| < 3$  possible outlier.
- Empirical rule** (bell-shaped): 68% within  $\mu \pm \sigma$ , 95% within  $\mu \pm 2\sigma$ , 99% within  $\mu \pm 3\sigma$ .
- Quartiles**: Q1 at position  $(n+1)/4$ ; Q3 at  $3(n+1)/4$ ; **IQR=Q3-Q1**.
- Box-plot fences**: inner =  $Q1 - 1.5 \cdot IQR$  &  $Q3 + 1.5 \cdot IQR$  ( $\rightarrow$  outlier); outer =  $\pm 3 \cdot IQR$  ( $\rightarrow$  serious outlier).

**ROBUSTNESS** **Sensitive to outliers**: mean, range,  $s/\sigma$ . **Resistant**: median, IQR. Bar chart = qualitative (gaps); histogram = quantitative (no gaps).

#### EX · BOX PLOT {30,40,70,81,90,118,123}

$Q1=70$ , median=81,  $Q3=90 \rightarrow IQR=20$ . Fences:  $70 - 1.5(20)=40$  &  $90 + 1.5(20)=120$ . Points outside  $\rightarrow$  **outliers: 30 & 123**; whiskers reach 40 & 118.

#### EX · MEAN VS. MEDIAN (N=9 COMPENSATIONS)

86.24, 88.77, 88.95, 94.62, **95.42**, 95.60, 95.64, 97.18, 99.80  $\rightarrow$  median = middle = **95.42**; mean = 93.58 (pulled down by the low value — mean < median here).

## §9 The Sample Midterm — Fully Worked

**The game**: Roll a fair die. **5 or 6**  $\rightarrow$  Jar 1 (15 red, 5 green). **1-4**  $\rightarrow$  Jar 2 (8 red, 12 green). So  $P(5,6)=2/6=1/3$ ,  $P(1-4)=4/6=2/3$ ;  $P(G|Jar1)=5/20=.25$ ,  $P(G|Jar2)=12/20=.6$ .

#### P1 · P(GREEN BALL) — TOTAL PROBABILITY

$$P(G)=P(G|Jar1)P(5,6)+P(G|Jar2)P(1-4) = .25(1/3)+.6(2/3) = .0833+.40 = \mathbf{.4833}$$

#### P2 · P(ROLLED 5-6 | GREEN) — BAYES

$$= P(G|5,6)P(5,6) / P(G) = .0833 / .4833 = \mathbf{.1724}$$

#### P3 · P(ROLLED 1-4 U GREEN)

$$= P(1-4)+P(G)-P(1-4 \cap G) = .6667+.4833-(.6 \cdot .6667) = .6667+.4833-.40 = \mathbf{.75}$$

#### P4 · DISTRIBUTION OF TOTAL PROFIT $Z=X+Y$

Die X: \$2 if 5-6, \$5 if 1-4. Ball Y: \$4 red, \$3 green.

| Outcome     | Z | P(Z)                    |
|-------------|---|-------------------------|
| 5,6 & Green | 5 | $1/3 \cdot .25 = .0833$ |
| 5,6 & Red   | 6 | $1/3 \cdot .75 = .25$   |
| 1-4 & Green | 8 | $2/3 \cdot .6 = .40$    |
| 1-4 & Red   | 9 | $2/3 \cdot .4 = .2667$  |

#### P5 · E(Z) AND $\Sigma_z$

$$E(Z)=5(.0833)+6(.25)+8(.40)+9(.2667)=\mathbf{7.517}$$

$$E(Z^2)=25(.0833)+36(.25)+64(.40)+81(.2667)=58.283$$

$$\text{Var}=58.283-7.517^2=1.783 \rightarrow \sigma_z=\mathbf{1.335}$$

#### P6 · E(XY)

$$= \Sigma(X \cdot Y)P = 6(.0833)+8(.25)+15(.40)+20(.2667) = .5+2+6+5.333 = \mathbf{13.833}$$

#### P7 · WIN \$9 $\geq$ 25× IN 100 PLAYS — BINOMIAL $\approx$ NORMAL

"Win \$9" = roll 1-4 & red,  $p=.2667$ .  $n=100$ : mean= $np=26.67$ ,  $\sigma=\sqrt{np(1-p)}=4.42$ .

Check  $n\pi, n(1-\pi) \geq 5$  ✓.  $z=(24.5-26.67)/4.42=-.49 \rightarrow P(X \geq 25)=P(Z > -.49) \approx \mathbf{.69}$  ( $1/2$  continuity correction)

#### P8 · VAR(2X) VS VAR(X+Y) · VAR=49 EACH, INDEP.

$$\text{Var}(2X)=2^2 \cdot 49=\mathbf{196}$$
.  $\text{Var}(X+Y)=49+49=\mathbf{98}$  ( $\text{Cov}=0$ ).  $\rightarrow$  **Var(2X) > Var(X+Y)**

#### P9 · INDEPENDENT? $P(A)=.5$ , $P(B)=.3$ , $P((A \cup B)^c)=.42$

If independent:  $P(A \cap B)=.15 \rightarrow P(A \cup B)=.65 \rightarrow P((A \cup B)^c)=.35$ . Given  $.42 \neq .35 \rightarrow$  **NOT independent**.

#### P10 · CORN CHIPS: $M=11.95$ , $\Sigma=22$ , $N=100$

(a) **P(total > 1194)**: sum mean= $100(11.95)=1195$ ,  $SD=\sqrt{100(.22)}=2.2$ .  $z=(1194-1195)/2.2=-.4545 \rightarrow P(Z > -.45)=\mathbf{.675}$

(b) **P( $\bar{x} < 11.97$ )**:  $SE=.22/\sqrt{100}=.022$ .  $z=(11.97-11.95)/.022=.909 \rightarrow \mathbf{.818}$

(c) " $n > 30 \Rightarrow$  individual bag is normal"? **No**. CLT normalizes the sample **mean/sum**, not an individual bag — it keeps the population shape for any  $n$ .

## More Practice · HW2 & Notes Problem Types

#### OVERBOOKING · EXPECTED VALUE OF A BINOMIAL FUNCTION

40 rooms, 50 reservations,  $P(\text{show})=.8 \rightarrow X \sim \text{Bin}(50, .8)$ . Pay \$150 per guest bumped ( $x > 40$ ).

$$E[\text{comp}] = 150 \cdot \sum_{x=41..50} (x-40) \cdot P(X=x) = 150(1.119) = \mathbf{\$167.78}$$

#### POISSON · "AT LEAST 3"

$$\lambda=2 \text{ in the window: } P(X \geq 3)=1-[P(0)+P(1)+P(2)]=1-e^{-2}(1+2+2)=1-.677=\mathbf{.323}$$

#### ACCEPTANCE SAMPLING · HYPERGEOMETRIC

Ship  $N=200$ , inspect 12, reject if **any** defective. With 10 defective:

$$P(\text{reject})=1-C(190,12)/C(200,12)=\mathbf{.469}$$
 (binomial approx  $1-(1-10/200)^{12} \approx .460$ ).

#### TWO-STAGE BAYES · COIN PICKS A JAR

Jar A 7R/3G, Jar B 15R/30G; fair coin ( $H \rightarrow A$ ,  $T \rightarrow B$ ).  $P(\text{red})=.5(.7)+.5(1/3)=\mathbf{.517}$ .

$$P(\text{tails}|\text{green})=[.5(2/3)]/[.5(.3)+.5(2/3)]=\mathbf{.690}$$

### STANDARDIZE TO COMPARE ACROSS CLASSES

Physics 76 ( $\mu=70, \sigma=5$ )  $\rightarrow z=1.2$ ; History 81 ( $80,3$ )  $\rightarrow .33$ ; Stats 78 ( $75,4$ )  $\rightarrow .75$ . Highest  $z \Rightarrow$

**Physics is best** relative to its class.

## §10 One-Glance Reference: Formulas, z-Table & Traps

### Master formula sheet

| Quantity            | Formula                                                                |
|---------------------|------------------------------------------------------------------------|
| Union               | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$                              |
| Conditional         | $P(A B) = P(A \cap B) / P(B)$                                          |
| Independent         | $P(A \cap B) = P(A)P(B)$                                               |
| Bayes               | $P(B A) = P(A B)P(B) / P(A)$                                           |
| Combinations        | $C(n,k) = n! / [k!(n-k)!]$                                             |
| $E(X)$              | $\sum x \cdot p(x)$                                                    |
| $\text{Var}(X)$     | $\sum x^2 p(x) - \mu^2$                                                |
| $\text{Var}(aX+b)$  | $a^2 \cdot \text{Var}(X)$                                              |
| $\text{Cov}(X,Y)$   | $E(XY) - \mu_X \mu_Y = \rho \sigma_X \sigma_Y$                         |
| $\text{Var}(aX+bY)$ | $a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \cdot \text{Cov}$         |
| Binomial            | $C(n,k) \pi^k (1-\pi)^{n-k}$ ; $\mu = n\pi$ , $\sigma^2 = n\pi(1-\pi)$ |
| Poisson             | $e^{-\lambda} \lambda^k / k!$ ; $\mu = \sigma^2 = \lambda$             |
| Hypergeom.          | $C(r,x)C(N-r,n-x) / C(N,n)$                                            |
| Standardize         | $z = (x - \mu) / \sigma$                                               |
| Sum of $n$          | mean $n\mu$ , SD $\sigma\sqrt{n}$                                      |
| Average of $n$      | mean $\mu$ , SE $\sigma/\sqrt{n}$                                      |
| Sample var          | $s^2 = \sum (x - \bar{x})^2 / (n-1)$                                   |

### Standard normal — $A(z) = P(0 \leq Z \leq z)$

| z    | A(z)  | z     | A(z)  | z    | A(z)  |
|------|-------|-------|-------|------|-------|
| 0.25 | .0987 | 1.00  | .3413 | 1.75 | .4599 |
| 0.50 | .1915 | 1.15  | .3749 | 1.96 | .4750 |
| 0.67 | .2486 | 1.28  | .3997 | 2.00 | .4772 |
| 0.75 | .2734 | 1.33  | .4082 | 2.33 | .4901 |
| 0.84 | .2995 | 1.50  | .4332 | 2.50 | .4938 |
| 0.90 | .3159 | 1.645 | .4500 | 3.00 | .4987 |

**Key criticals:**  $z=1.28 \rightarrow .40$  (90th pct) ·  $1.645 \rightarrow .45$  (95th) ·  $1.96 \rightarrow .475$  (97.5th) ·  $2.33 \rightarrow .49$  (99th).  $P(Z \leq z) = .5 + A(z)$ .

### Excel functions

- =BINOM.DIST(k,n,p,FALSE/TRUE)** — binomial  $P(X=k) / P(X \leq k)$
- =POISSON.DIST(k,λ,FALSE/TRUE)**
- =NORM.S.DIST(z,TRUE)** —  $\Phi(z) = P(Z \leq z)$  (full cumulative)
- =NORM.DIST(x,μ,σ,TRUE)** · **=COMBIN(n,k)** · **=FACT(n)**

### Top exam traps & final checklist

- Variances add, SDs don't.** Never add standard deviations.
- $\text{Var}(aX) = a^2 \text{Var}(X)$**  — square the constant; +b does nothing to variance.
- Independent  $\Rightarrow \text{Cov}=0$ ;** if a  $\rho$  or  $\text{Cov}$  is given, you **MUST** include 2· $\text{Cov}$ .
- With replacement  $\rightarrow$  Binomial; without  $\rightarrow$  Hypergeometric.**
- Poisson  $\lambda$  = rate  $\times$  the exact time window asked.**
- CLT:** sum uses  $\sigma\sqrt{n}$ , average uses  $\sigma/\sqrt{n}$  — and it never makes an *individual* value normal.
- Sample  $s^2$  divides by  $n-1$ ;** population  $\sigma^2$  divides by  $N$ .
- This class' z-table is "0  $\rightarrow$  z"** — add .5 for a full left tail; use symmetry for negatives.
- Always draw the bell & shade** before computing a normal probability.
- "At least one" =  $1 - P(\text{none})$ .**